Time Clusters of Alzheimer’s Disease Progression

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1) Project Description
2) Dataset
Agenda
3) Processing [preprocess & model proc.]
4) Modeling Results
5) Discovery Phase
6) Conclusions

Objective: Background

Issue

- Over 55 million adults worldwide are affected by Alzheimer's/dementia

Current Research Limitations

- Heavy reliance on probability models and simulated datasets
- Biomarker Focus
- No intuitive tools for progression

Objective: Goal

Predictive Modeling

Apply conventional modeling methods to predict binary cognitive status
Identify predictive features through multimodal modeling.

Advanced Clustering

Use clustering techniques to identify cognitive stages and transition phases suitable for clinical interpretation.

Cluster Deconstruction

Break down derived clusters to determine key features driving each state, informing future research and clinical modeling.

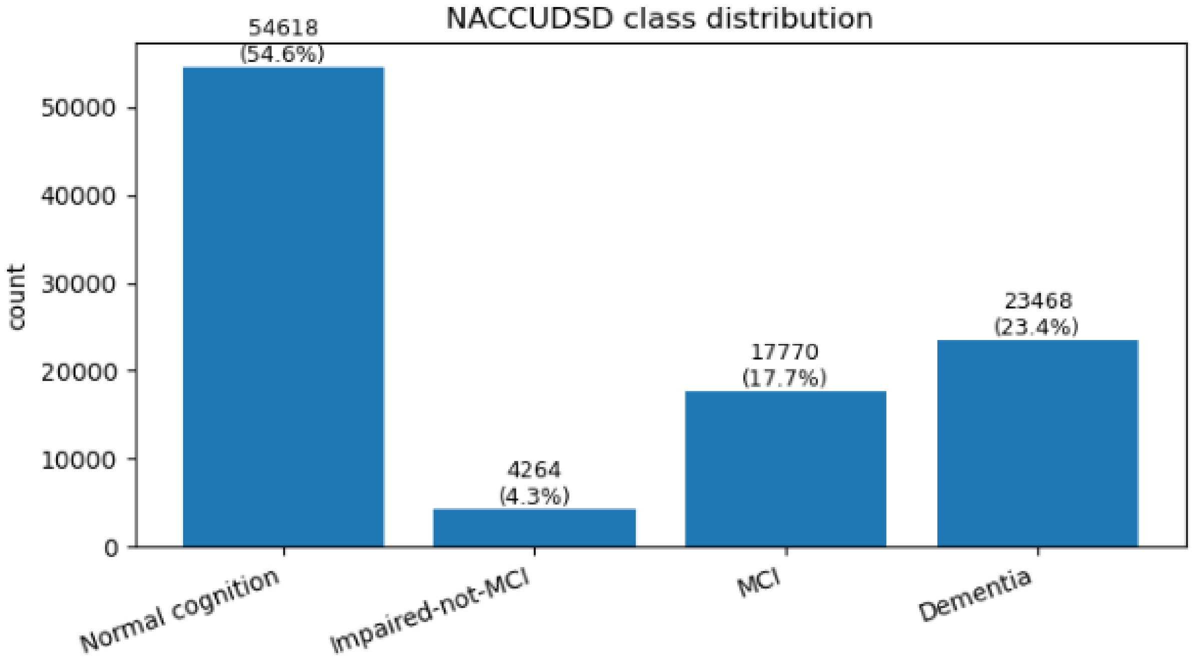
Dataset

- Longitudinal NACC_UDS
- Geographically diverse
- 2005-2025 years
- 54K+ subjects
- 3.7 average years of data per subject
- 1K+ number of variables
- 200K+ observation

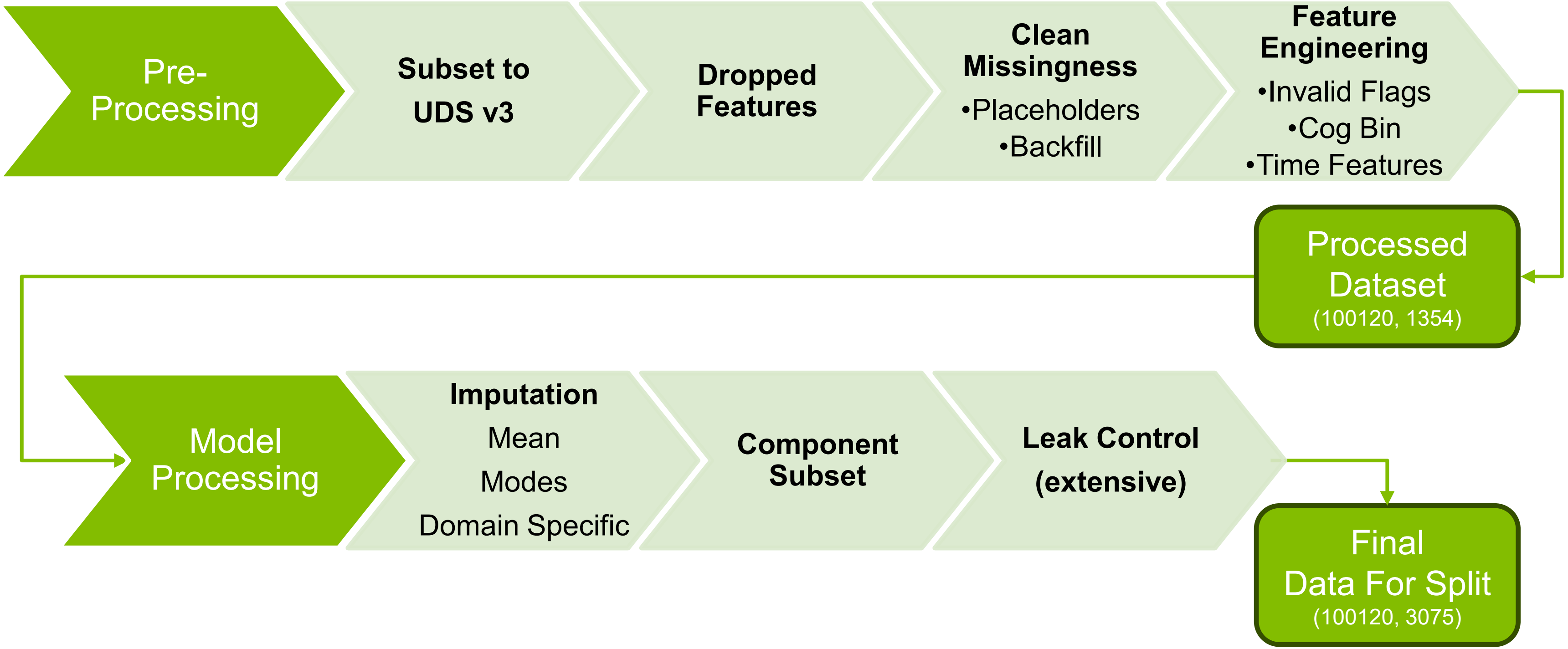
	Normal	Impaired Not MCI	MCI	Demented	Overall
N (rows)	54618	4264	17770	23468	100120
MOCA mean (SD)	25.37 (5.86)	23.67 (5.39)	21.81 (5.08)	12.93 (7.69)	22.19 (7.75)
Cog. Stat mean (SD)	6.80 (4.32)	7.41 (4.56)	7.47 (4.46)	7.31 (4.31)	7.06 (4.36)
BIRTHYR median	1946	1945	1944	1944	1945
RACE = 1 n (%)	43422 (79.5%)	3008 (70.5%)	13749 (77.4%)	20566 (87.6%)	80745 (80.6%)
SEX = 2 n (%)	35497 (65.0%)	2476 (58.1%)	9153 (51.5%)	11655 (49.7%)	58781 (58.7%)
EDUC = 18 n (%)	14951 (27.4%)	975 (22.9%)	3979 (22.4%)	4509 (19.2%)	24414 (24.4%)

Dataset

- Half data set normal
- Half impaired in somewhat
- Binning allows accuracy as primary evaluator for downstream modeling



Processing Pipeline Graphic



Model Processing: Temporal Split

Final Processed Data Set

Training data
subjects: 16002 | rows: 39290
All subjects’ rows before cutoff

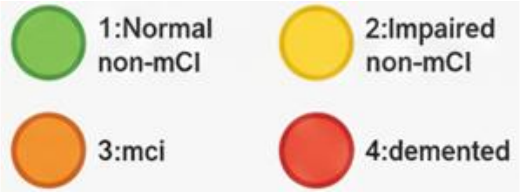
Test Data
subjects: 15893 | rows: 60830
All subjects’ rows after cutoff

1/1/2023

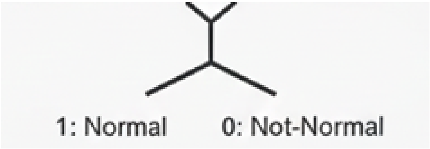
Modeling Results: Modeling Framework

What We predicted (Target Variables)

NACCUDSD (Discovery)



naccudud_bin (supervised)



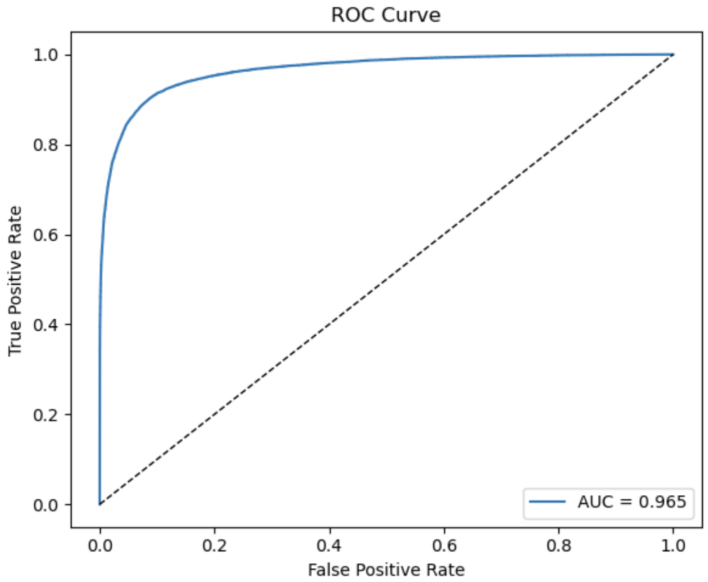
Modeling Approach

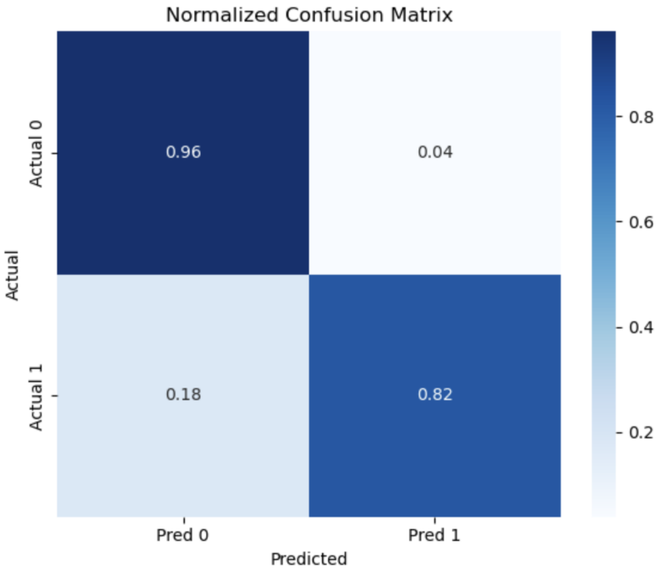
How We Predicted (Model Families)

- Logistic Regression (Interpretable)
- XGBoost (Non-linear, high perform.)
- GEE (Longitudinal performance)
- Hidden Markov Model (Latent time features)

Supervised Modeling: Logistic Regression

Cross-Validation: 5-fold subject level cross validation (GroupFold)
Loss Function: log-likelihood (corresponds to binary cross entropy loss)





Supervised Modeling: XGBOOST

Description of Model: Tree ensemble with boosting process that identifies complex patterns

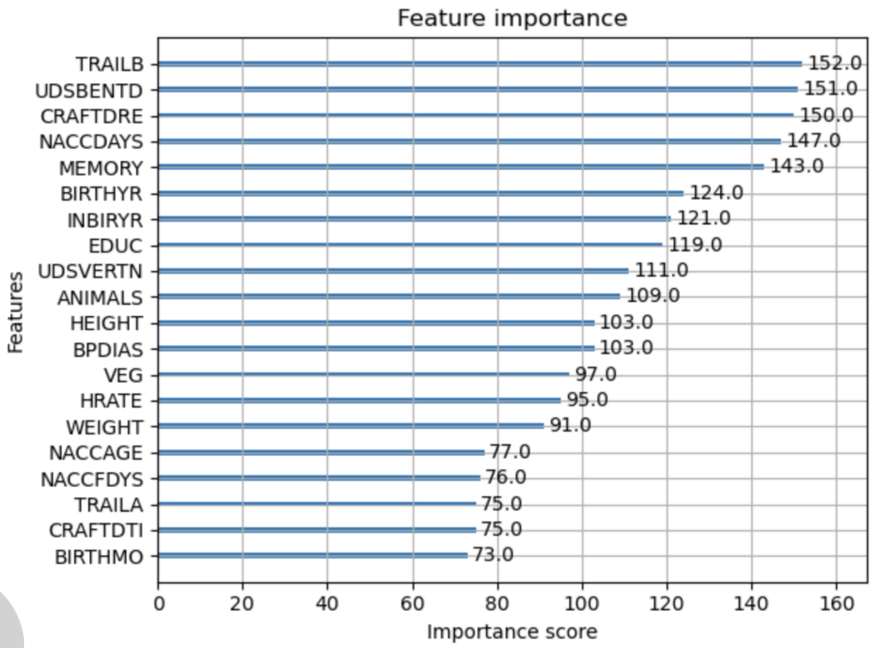
Why Use: Works well with messy high dimensional data with mixed variables and scale differences

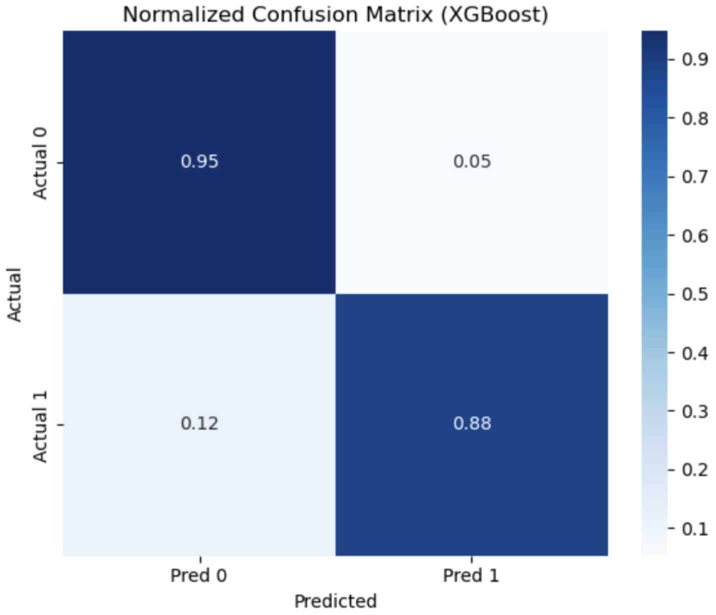
Model Type	# Features	Train Rows	Test Rows	Accuracy	Precision	Recall	F1
Lagged feature model	304 lag/temporal features	23,288	44,994	0.892	0.859	0.808	0.833
Full feature model	496 engineered features (no lags)	39,290	60,830	0.924	0.898	0.881	0.890

Tuning:
Lag versus no lag

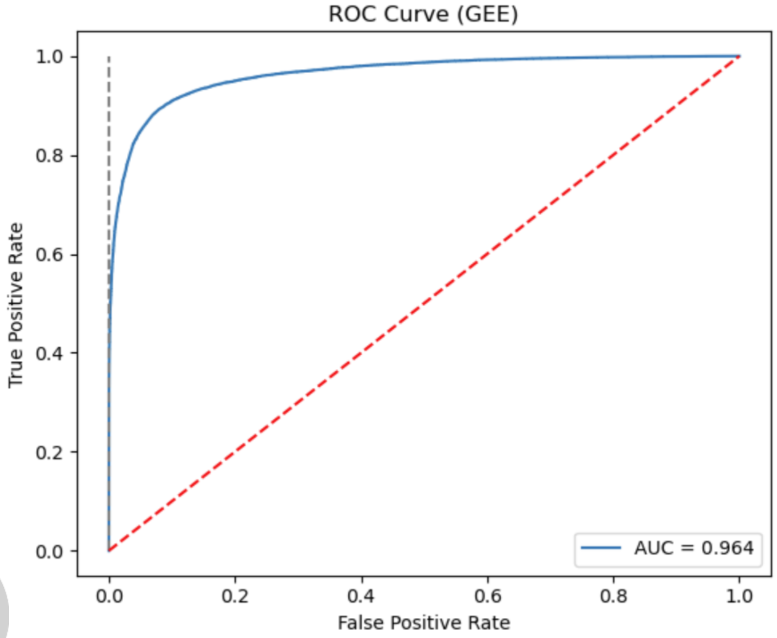
Loss Function:
logistic loss/binary cross-entropy

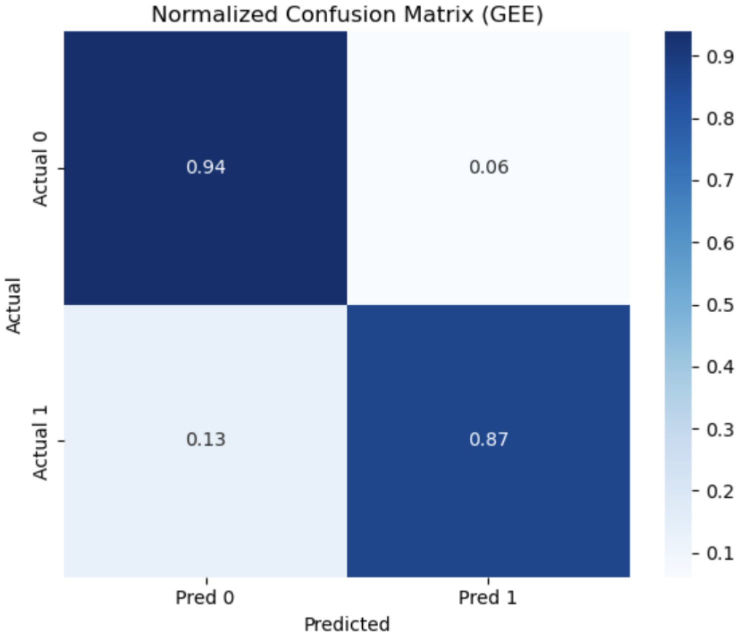
Supervised Modeling: Full XGBOOST





Supervised Modeling: GEE





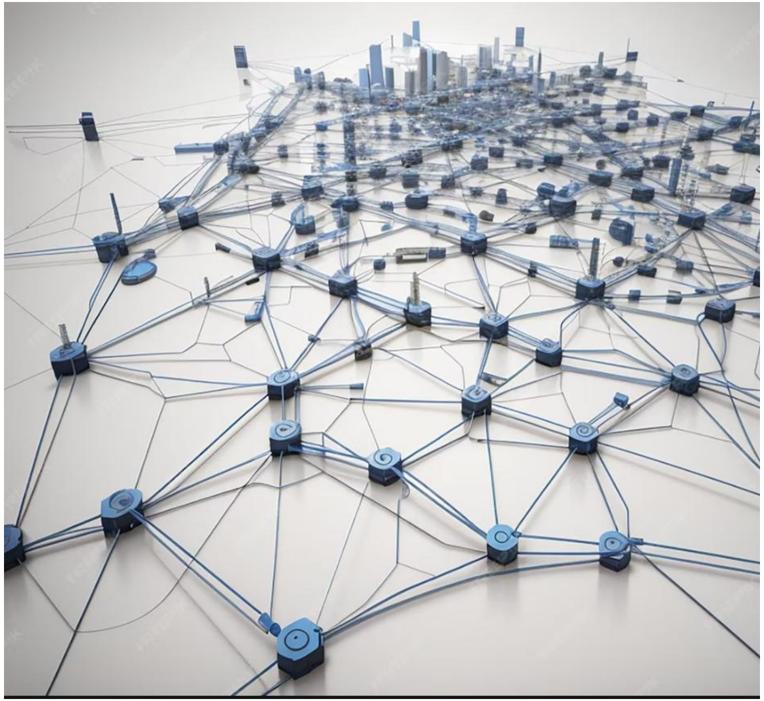
Supervised Modeling: HMM

Description of Model: *“HMMs are statistical frameworks designed to represent a Markov process with hidden, unobservable states”* [6]

Discovers hidden states by uncovering patterns and relationships in the observed data that are not directly visible from the observations alone.

Uses these latent states to model how individuals move from one cognitive stage to the next over time.

Project Use-case: Prediction and Discovery



Supervised Modeling: HMM

- A sequence of hidden states that follow a Markov process.
- Future state depends only on the current state, not on past states
- Each hidden state produces an emission according to known prob dist.

Likelihoods (Forward) > decode States (Viterbi) > get parameters from data (Baum–Welch).

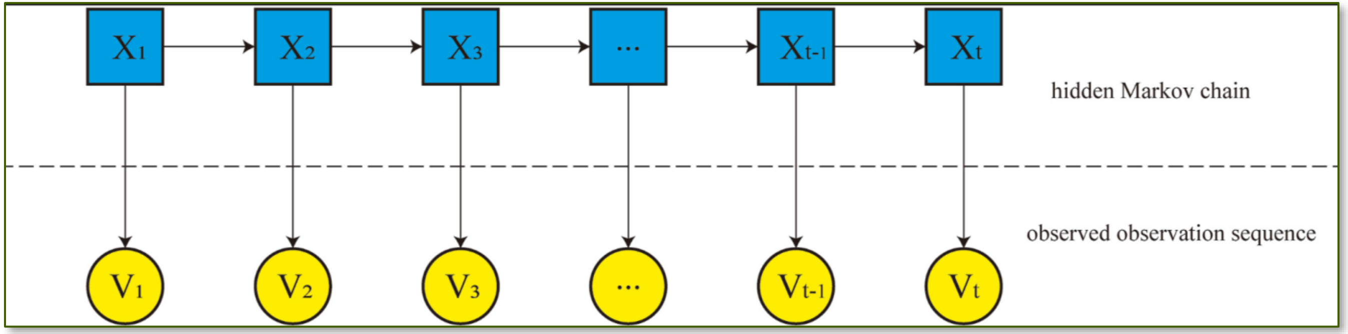


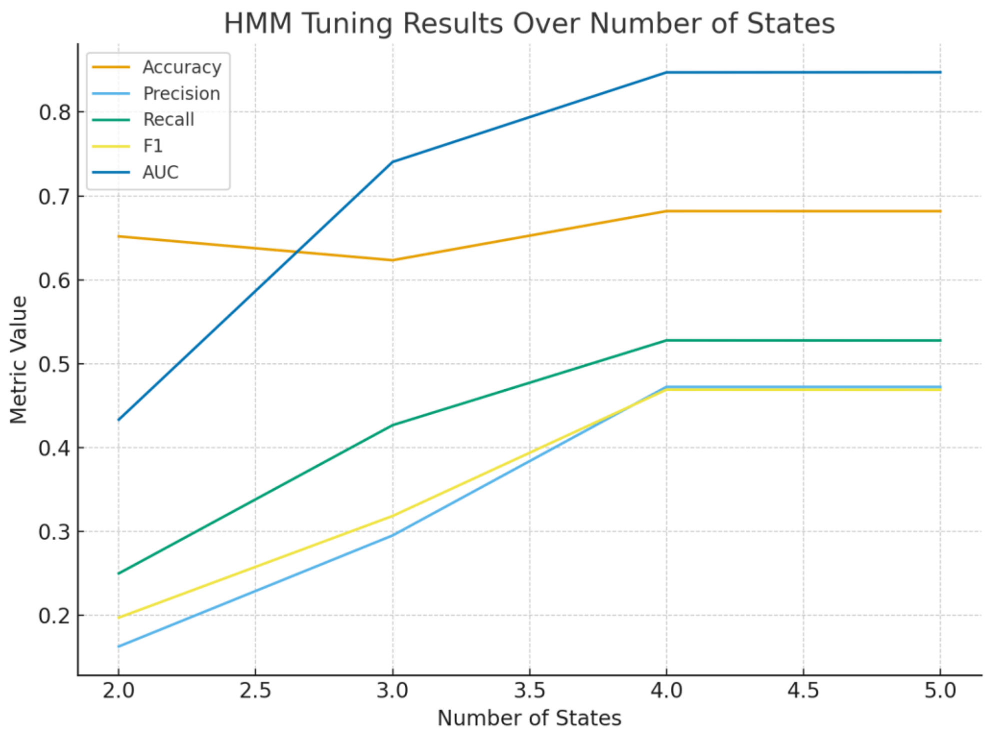
Image from Ma et. al demonstrating stochastic process in HMM which generates latent states from observed data

Supervised Modeling: HMM

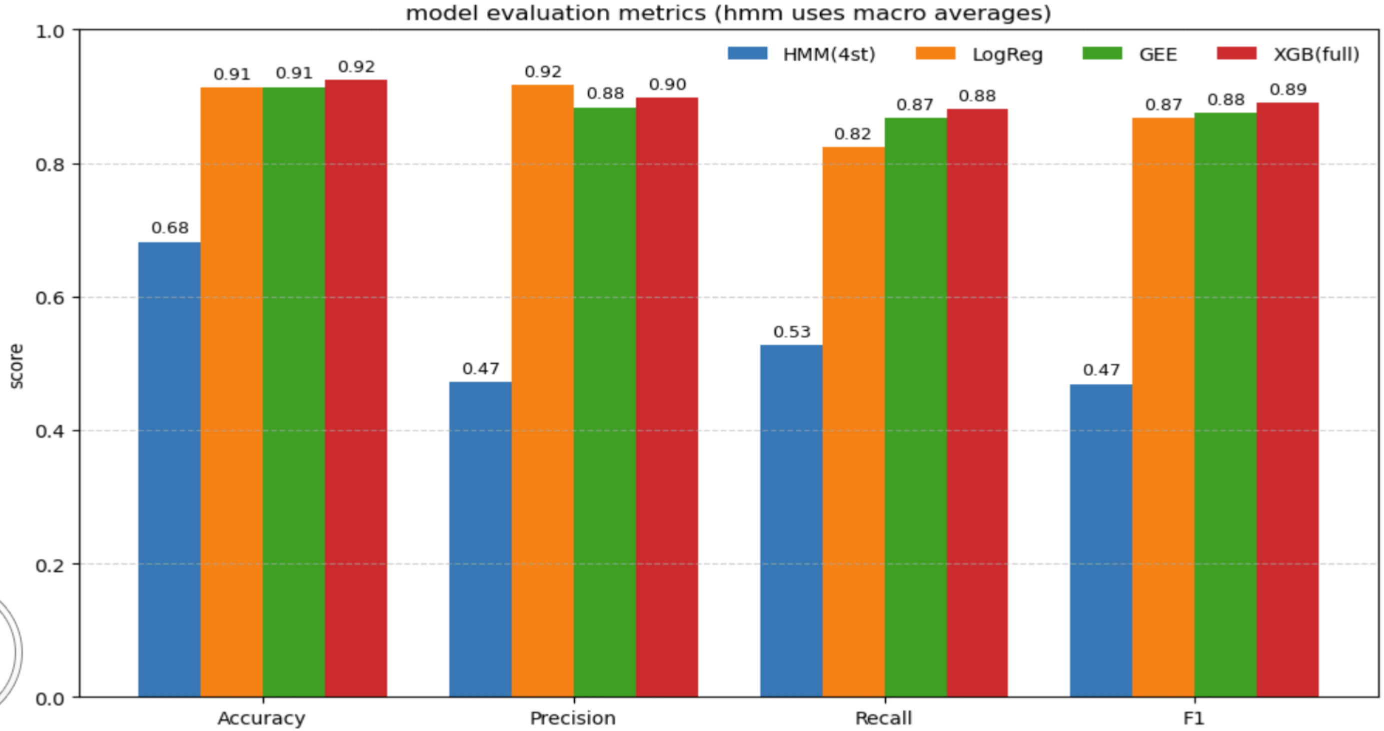
Tuning Parameter: Number of latent states

Performance plateau: Metrics stabilize around 4 states

Decision: Use the 4-state model for downstream discovery



Modeling Results: Model Comparison



Accuracy was best for XGBoost and consistently good throughout model metrics

HMM did not perform well but was anticipated

Discovery Phase: Unsupervised HMM

1. Our unsupervised HMM was expected to discover hidden cognitive stages people pass through over time.

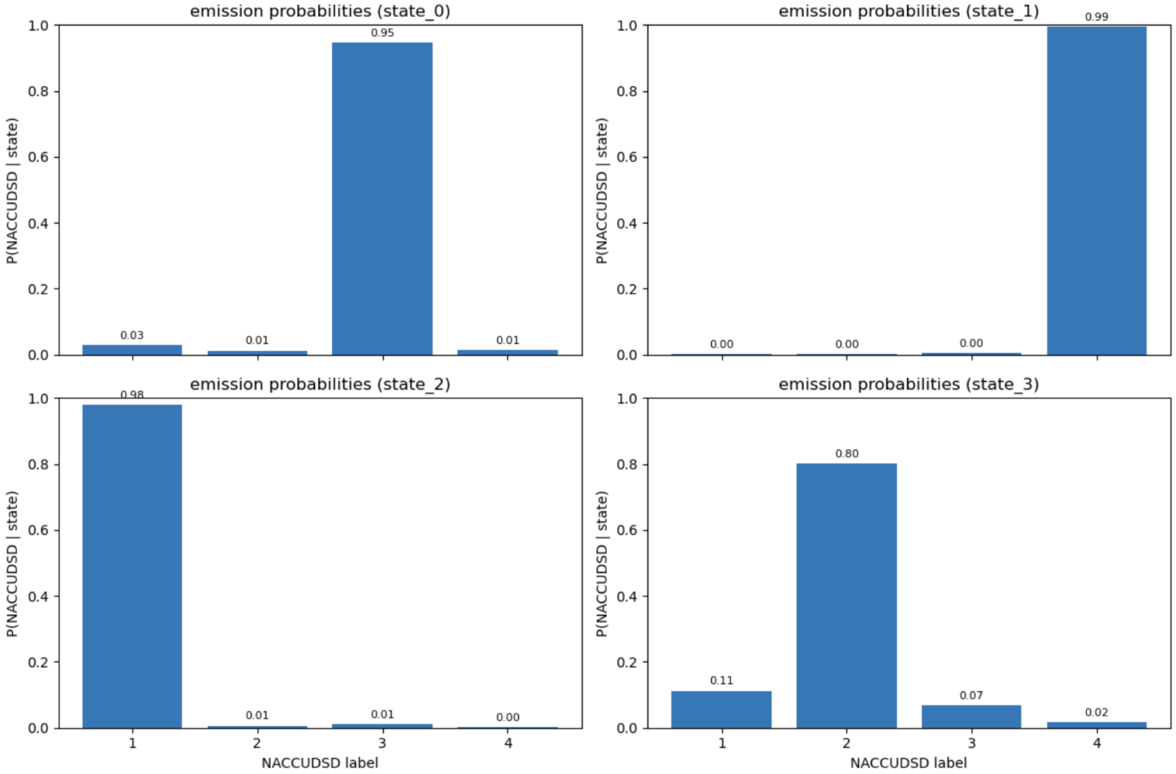
2. HMM finds invisible cognitive progression stages using only the observed diagnosis labels.

3. The model only sees what we observe at each visit and tries to infer the underlying progression.

Discovery Phase: Latent States

Emissions: the observed diagnosis at each visit (NACCUDSD: 1: normal, 2:Impaired Not MCI, 3:MCI, 4:dementia).

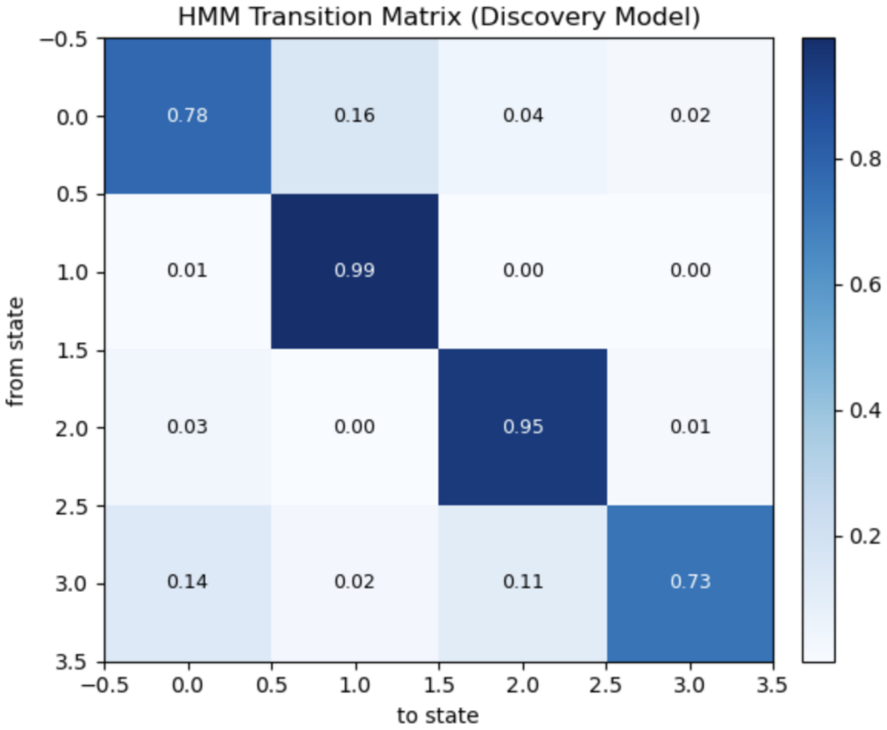
Distinct Cognitive States (0,1,2, 3) defined by upstream optimization



Discovery Phase: Latent States

Transition patterns between latent states

- Highly Stable States:1,2
- More fluid: 0,3



Discovery Phase: Interpetation

Based on emission patterns and transition dynamics, we see the clinical states as follows:

State 2: Normal/early → State 3: Mild Impairment → State 0: Moderate Impairment → State 1: Severe Impairment

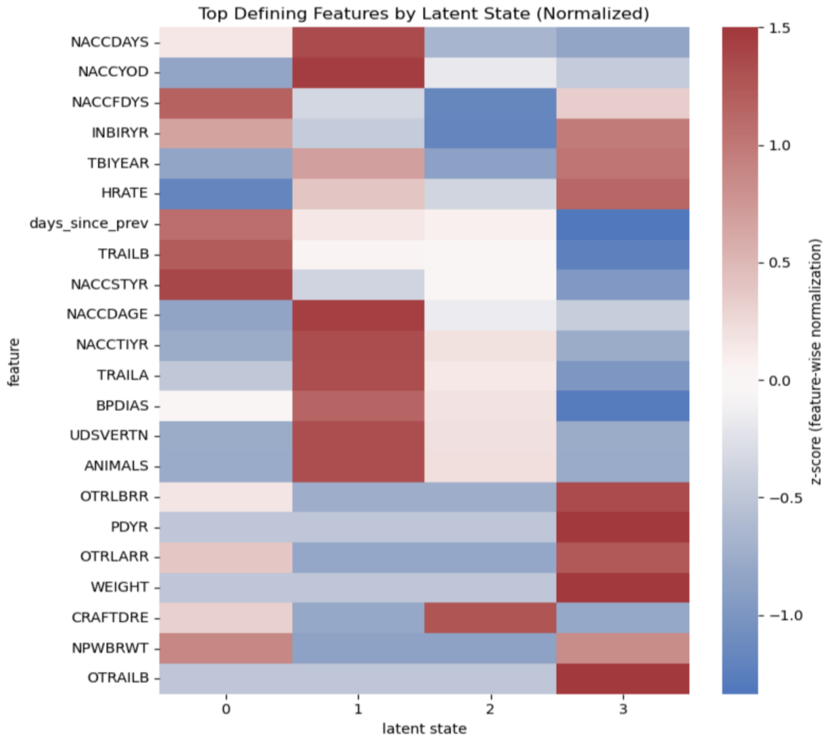
Discovery Phase: State Features

Normal/Early (state 2):
- Low cardiovascular markers
- healthy baselines

Mild Impairment (state 3):
- Mild decreases in heart health & cognition

Moderate Impairment (state 0):
- Increases in heart signals, functional declines and distinct declines in cognition

Sever Impairment (state 1):
- Largest shift away from mean for many features.



Discovery Phase: Implications

HMM Accomplishments:

- The HMM successfully discovered a clinically meaningful progression pathway without labels
- Revealed the feature patterns that define each stage

Clinical Implications:

We can detect and track cognitive decline trajectories automatically from real-world data, supporting earlier and more precise decision-making

Conclusions

- Classic predictive models performed well for identifying cognitive status
- Clinical diagnosis fields upstream removal ensured clinical characters drove modeling
- Unsupervised HMM revealed meaningful latent transition phases
- These findings lay the groundwork for automated tools that infer cognitive stage at any clinical visit.

Future Work

- Conduct domain-expert review of leak-mitigation strategies; high accuracy suggests possible residual leakage from correlated features.
- After resolving leakage concerns, develop a real-time clinical decision-support tool using the most informative cognitive and functional features

Citation

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